Devops and Cloud Assignment

Mini-Project : 4

Infrastructure Automation with Ansible and Terraform

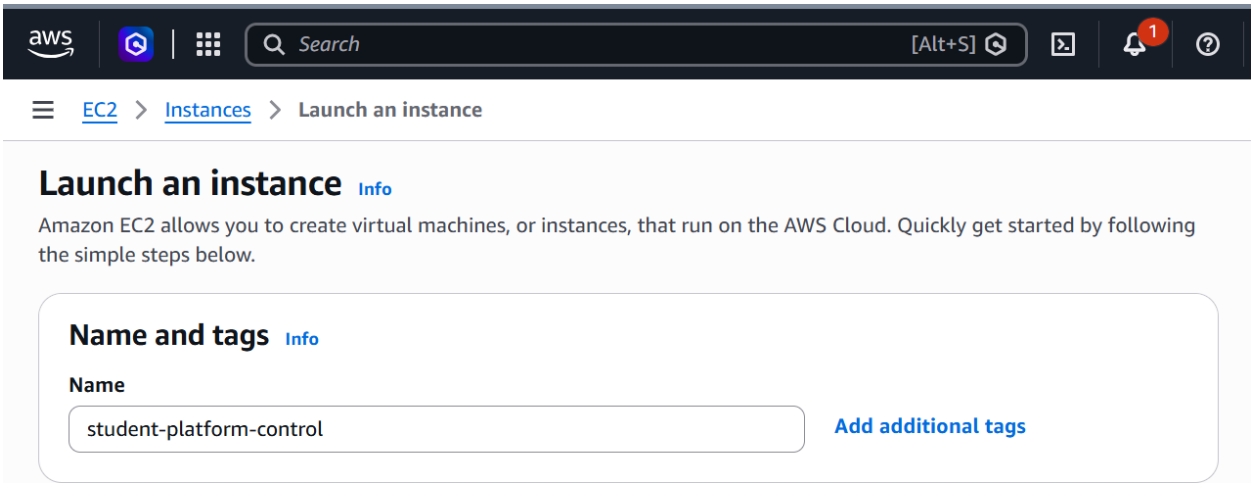
Enterprise Student Record and Academic Management Platform

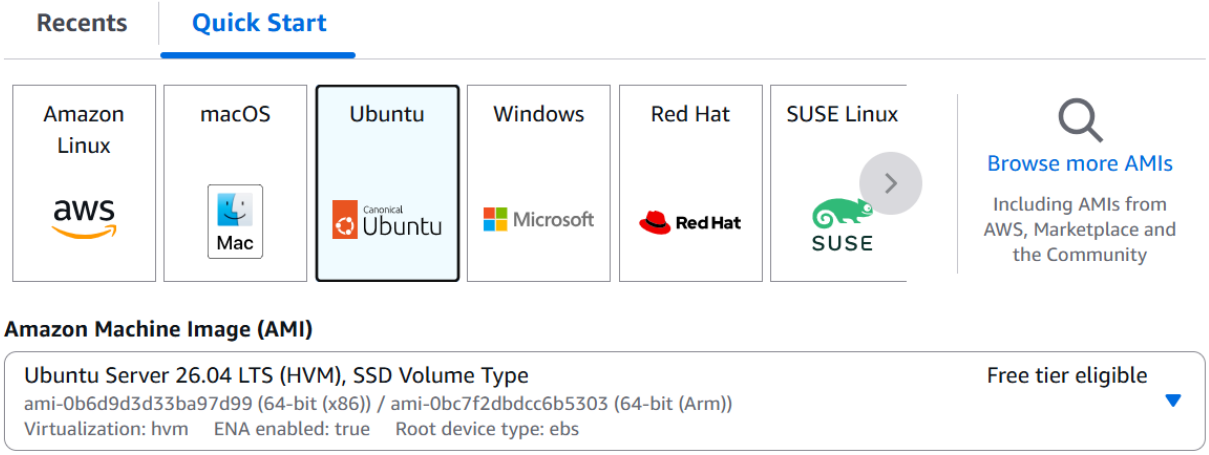
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PRN: 2125UDSF1093

Dept: SY-AIDS

Prepare your AWS and MobaXterm environment





Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

student_platform_key

Create new key pair

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

(•) Create security group ( ) Select existing security group

We'll create a new security group called '**launch-wizard-3**' with the following rules:

[x] Allow SSH traffic from
Helps you connect to your instance
My IP
136.233.205.186/32

[ ] Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

[ ] Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

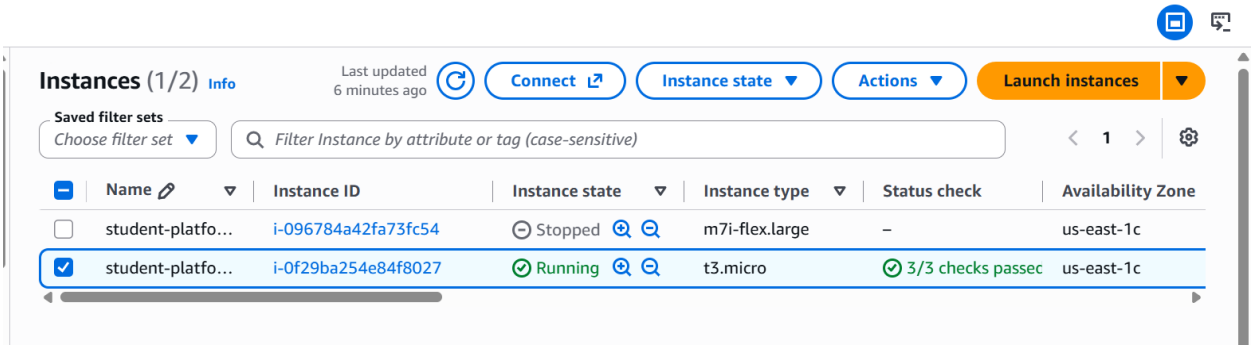
Configure storage Info

Advanced

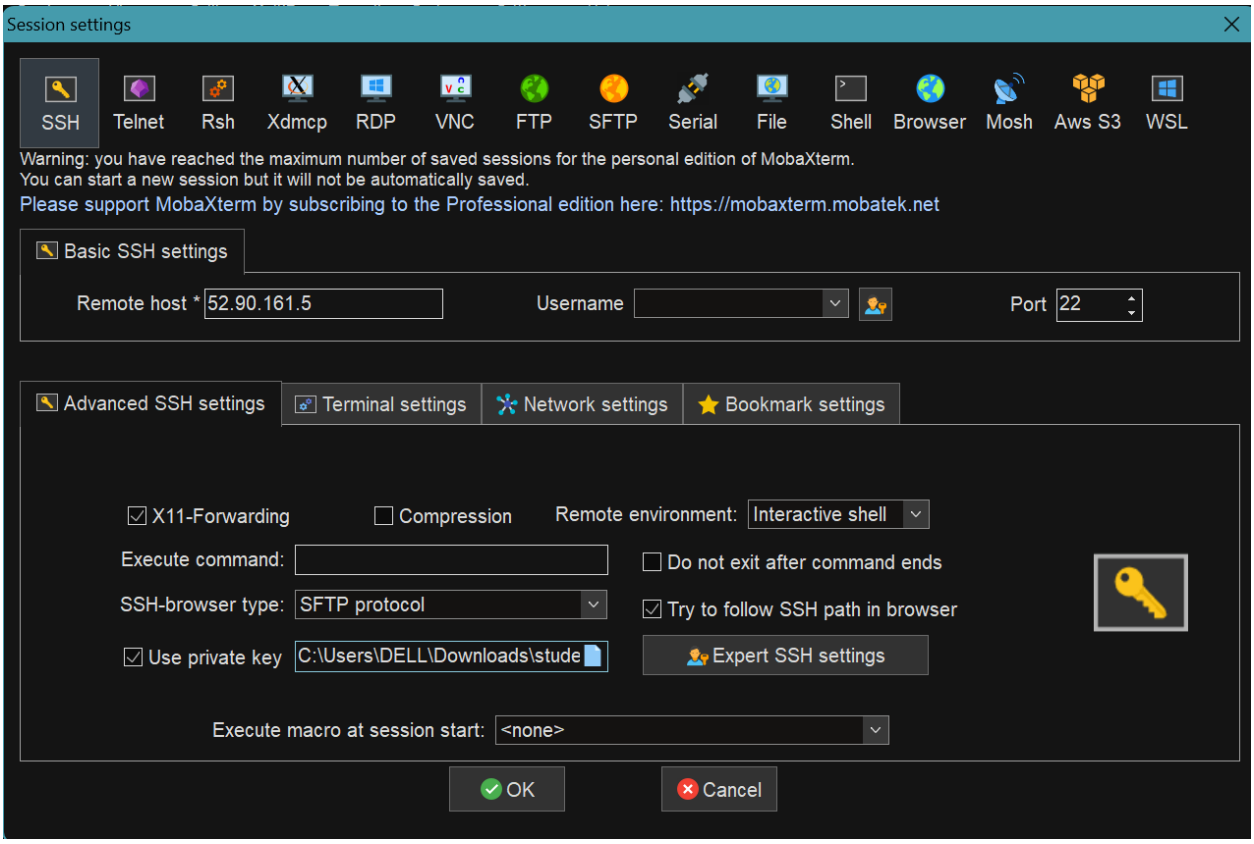
1x 20 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance



Connect through MobaXterm




```
ubuntu@ip-172-31-17-50:~$ whoami
ubuntu
ubuntu@ip-172-31-17-50:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-pol
icy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
ubuntu@ip-172-31-17-50:~$ pwd
/home/ubuntu
```

Install the prerequisite packages

```
ubuntu@ip-172-31-17-50:~$
sudo apt-get update
sudo apt-get install -y curl wget unzip jq git ca-certificates gnupg software-properties-common python3 python3-pip python3-venv
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [738 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [173 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [329 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [107 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [602 kB]
Fetched 2224 kB in 1s (2232 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.18.0-1ubuntu2.7).
curl set to manually installed.
wget is already the newest version (1.25.0-2ubuntu4.4).
wget set to manually installed.
jq is already the newest version (1.8.1-4ubuntu2).
jq set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
```

Install AWS CLI v2

```
ubuntu@ip-172-31-17-50:~$ cd ~
ubuntu@ip-172-31-17-50:~$ curl -fsSL "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o awscliv2.zip
ubuntu@ip-172-31-17-50:~$ unzip -q awscliv2.zip
replace aws/THIRD_PARTY_LICENSES? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
ubuntu@ip-172-31-17-50:~$ sudo ./aws/install --update
Found same AWS CLI version: /usr/local/aws-cli/v2/2.38.0. Skipping install.
ubuntu@ip-172-31-17-50:~$ aws --version
aws-cli/2.38.0 Python/3.14.6 Linux/7.0.0-1006-aws exe/x86_64.ubuntu.26
ubuntu@ip-172-31-17-50:~$
```

Install Terraform

```
ubuntu@ip-172-31-17-50:~$ sudo apt-get install -y terraform
terraform -version
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 50 not upgraded.
Need to get 35.9 MB of archives.
After this operation, 121 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com resolute/main amd64 terraform amd64 1.16.6-1 [35.9 MB]
Fetched 35.9 MB in 0s (99.1 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 131725 files and directories currently installed.)
Preparing to unpack .../terraform_1.16.6-1_amd64.deb ...
Unpacking terraform (1.16.6-1) ...
Setting up terraform (1.16.6-1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  7.0.0-1006-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 7.0.0-1014-aws.
```

```
ubuntu@ip-172-31-17-50:~$ terraform -version
Terraform v1.16.6
on linux_amd64
ubuntu@ip-172-31-17-50:~$
```

Install Ansible

```
ubuntu@ip-172-31-17-50:~$ sudo add-apt-repository --yes --update ppa:ansible/ansible
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/
Suites: resolute
Components: main
'
Description:
Ansible is a radically simple IT automation platform that makes your applications and systems easier to deploy. Avoid writing scripts or custom code to
deploy and update your applications— automate in a language that approaches plain English, using SSH, with no agents to install on remote systems.

http://ansible.com/

If you face any issues while installing Ansible PPA, file an issue here:
https://github.com/ansible-community/ppa/issues
More info: https://launchpad.net/~ansible/+archive/ubuntu/ansible
Adding repository.
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://apt.releases.hashicorp.com resolute InRelease
Get:6 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu resolute InRelease [20.0 kB]
Get:7 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu resolute/main amd64v3 Packages [940 B]
Get:8 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu resolute/main Translation-en [472 B]
Fetched 21.4 kB in 1s (31.1 kB/s)
```

```
ubuntu@ip-172-31-17-50:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (14.5.0-1ppa~resolute).
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 50 not upgraded.
ubuntu@ip-172-31-17-50:~$ ansible --version
ansible [core 2.21.5]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.14.4 (main, Aug 20 2026, 10:41:58) [GCC 15.2.0] (/usr/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
ubuntu@ip-172-31-17-50:~$
```

Install Docker

```
ubuntu@ip-172-31-17-50:~$ sudo apt install docker.io
Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown    cgroupfs-mount  debootstrap    docker-compose-v2  rinse        zfs-fuse
  aufs-tools  | cgroup-lite   docker-buildx  docker-doc         rootlesskit  | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 50
  Download size: 74.0 MB
  Space needed: 281 MB / 14.9 GB available

Continue? [Y/n] Y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.1 [28.1 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dns-root-data all 2025080400build1 [6022 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 dnsmasq-base amd64 2.92-1ubuntu0.5 [441 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 docker.io amd64 29.1.3-0ubuntu4.1 [35.5 MB]
```

```
ubuntu@ip-172-31-17-50:~$ sudo systemctl enable --now docker
sudo docker run --rm hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

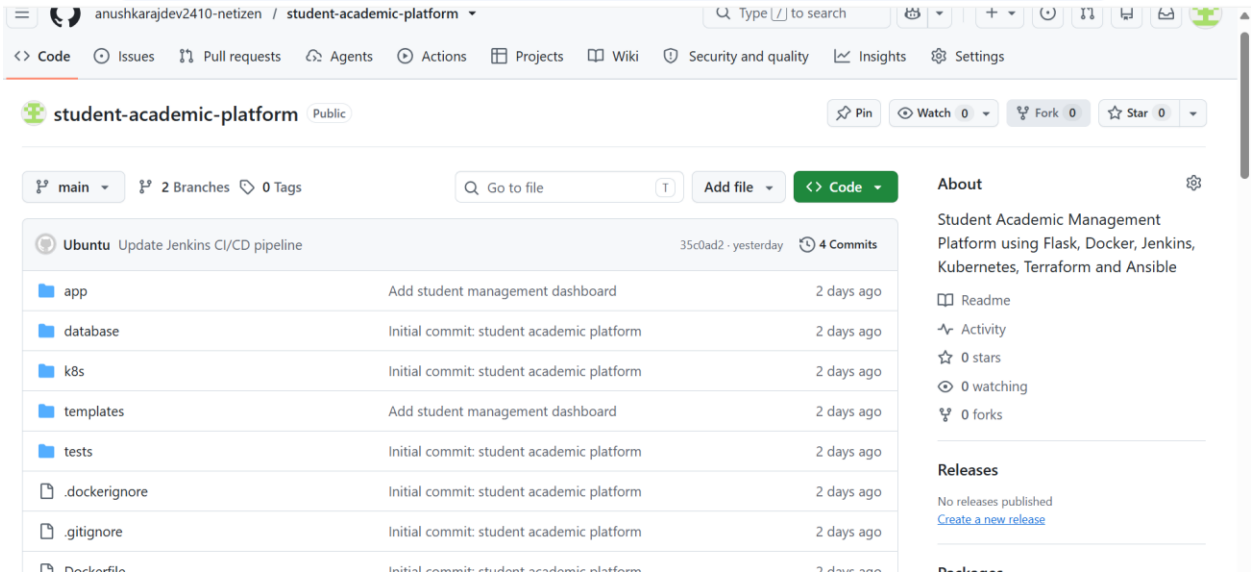
For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Verify all tools

```
ubuntu@ip-172-31-17-50:~$ for tool in aws terraform ansible docker git python3; do
  if command -v "$tool" >/dev/null 2>&1; then
    echo "$tool: INSTALLED"
  else
    echo "$tool: NOT INSTALLED"
  fi
done
aws: INSTALLED
terraform: INSTALLED
ansible: INSTALLED
docker: INSTALLED
git: INSTALLED
python3: INSTALLED
ubuntu@ip-172-31-17-50:~$
```

Project File:

Reopsitory link: https://github.com/anushkarajdev2410-netizen/student-academic-platform.git



Check existing project and tools

```
ubuntu@ip-172-31-17-50:~$ for tool in aws terraform ansible docker git python3; do
  if command -v "$tool" >/dev/null 2>&1; then
    echo "$tool: INSTALLED"
  else
    echo "$tool: NOT INSTALLED"
  fi
done
aws: INSTALLED
terraform: INSTALLED
ansible: INSTALLED
docker: INSTALLED
git: INSTALLED
python3: INSTALLED
ubuntu@ip-172-31-17-50:~$
# 1. Check installed tools
for tool in aws terraform ansible docker git python3; do
  if command -v "$tool" >/dev/null 2>&1; then
    echo "$tool: INSTALLED"
  else
    echo "$tool: NOT INSTALLED"
  fi
done

# 2. Check the Terraform repository filename
ls -l /etc/apt/sources.list.d/

# 3. Check your Ubuntu version
. /etc/os-release
echo "$PRETTY_NAME ($VERSION_CODENAME)"
```

Download project

```
ubuntu@ip-172-31-17-50:~$ cd ~
git clone https://github.com/anushkarajdev2410-netizen/student-academic-platform.git
cd student-academic-platform
Cloning into 'student-academic-platform'...
remote: Enumerating objects: 36, done.
remote: Counting objects: 100% (36/36), done.
remote: Compressing objects: 100% (30/30), done.
remote: Total 36 (delta 6), reused 33 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (36/36), 16.45 KiB | 1.83 MiB/s, done.
Resolving deltas: 100% (6/6), done.
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform
git status
git remote -v
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
origin  https://github.com/anushkarajdev2410-netizen/student-academic-platform.git (fetch)
origin  https://github.com/anushkarajdev2410-netizen/student-academic-platform.git (push)
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

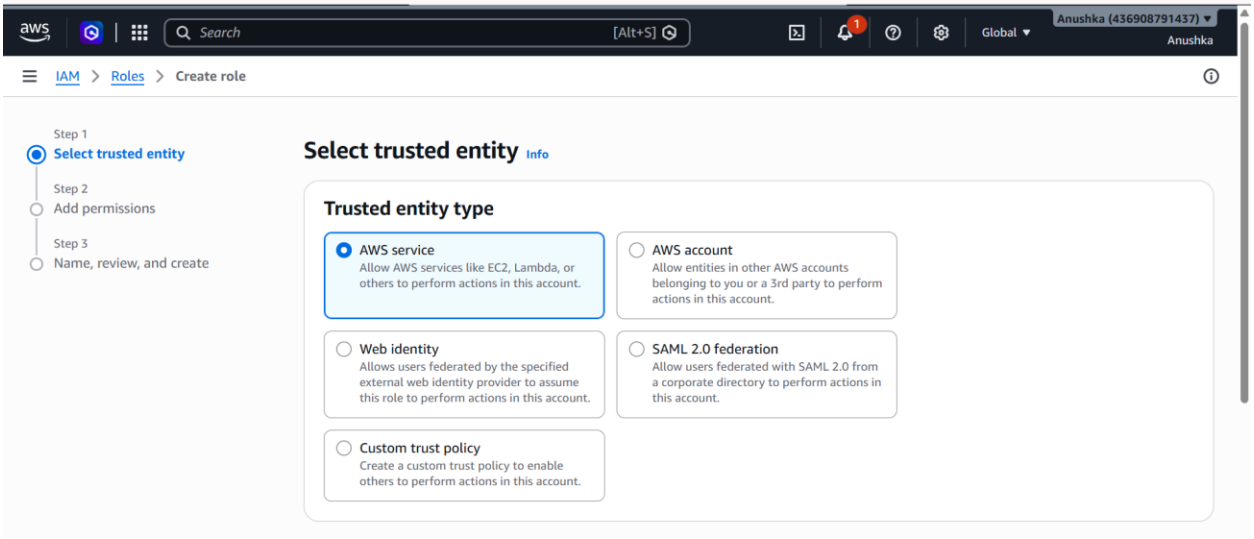
Inspect the project structure

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ pwd
ls -la
find . -maxdepth 3 -type f \
  -not -path './.git/*' \
  -not -path './.venv/*' | sort | head -200
/home/ubuntu/student-academic-platform
total 60
drwxrwxr-x 8 ubuntu ubuntu 4096 Oct 14 10:33 .
drwxr-x--- 8 ubuntu ubuntu 4096 Oct 14 10:33 ..
-rw-rw-r-- 1 ubuntu ubuntu   75 Oct 14 10:33 .dockerignore
drwxrwxr-x 7 ubuntu ubuntu 4096 Oct 14 10:35 .git
-rw-rw-r-- 1 ubuntu ubuntu  275 Oct 14 10:33 .gitignore
-rw-rw-r-- 1 ubuntu ubuntu  303 Oct 14 10:33 Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 1543 Oct 14 10:33 Jenkinsfile
-rw-rw-r-- 1 ubuntu ubuntu 1164 Oct 14 10:33 README.md
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 14 10:33 app
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 database
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 k8s
-rw-rw-r-- 1 ubuntu ubuntu   58 Oct 14 10:33 requirements.txt
-rw-rw-r-- 1 ubuntu ubuntu  114 Oct 14 10:33 run.py
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 templates
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 tests
./.dockerignore
./.gitignore
./Dockerfile
./Jenkinsfile
./README.md
./app/__init__.py
./app/db.py
./app/routes.py
./app/services.py
```

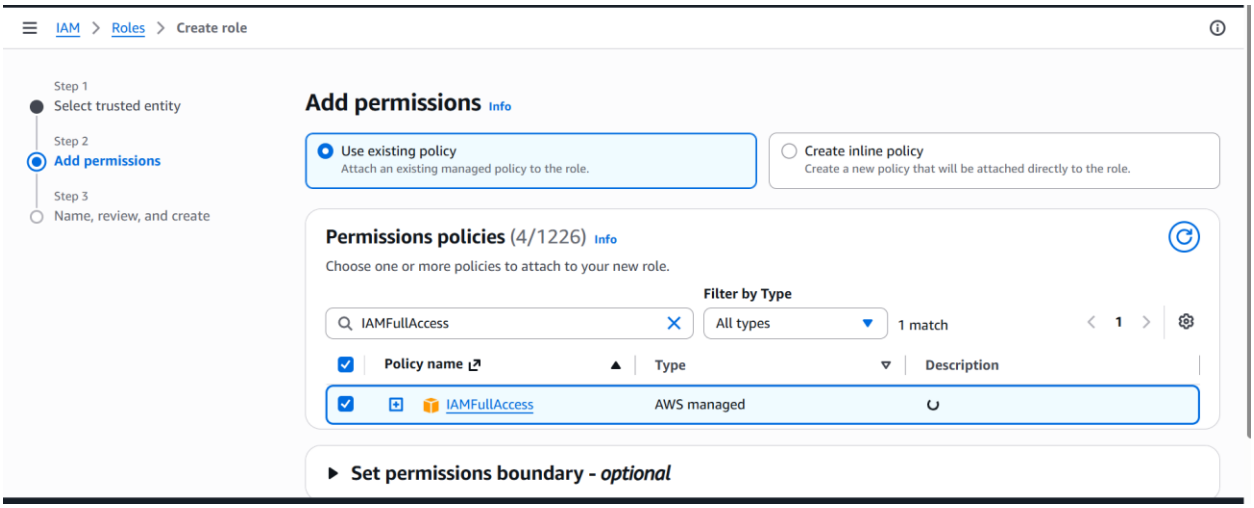
Check whether Terraform is ready

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ terraform -version
Terraform v1.16.6
on linux_amd64
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Create IAM Role



Attach all the required policies



Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

student-academic-ec2-role

Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

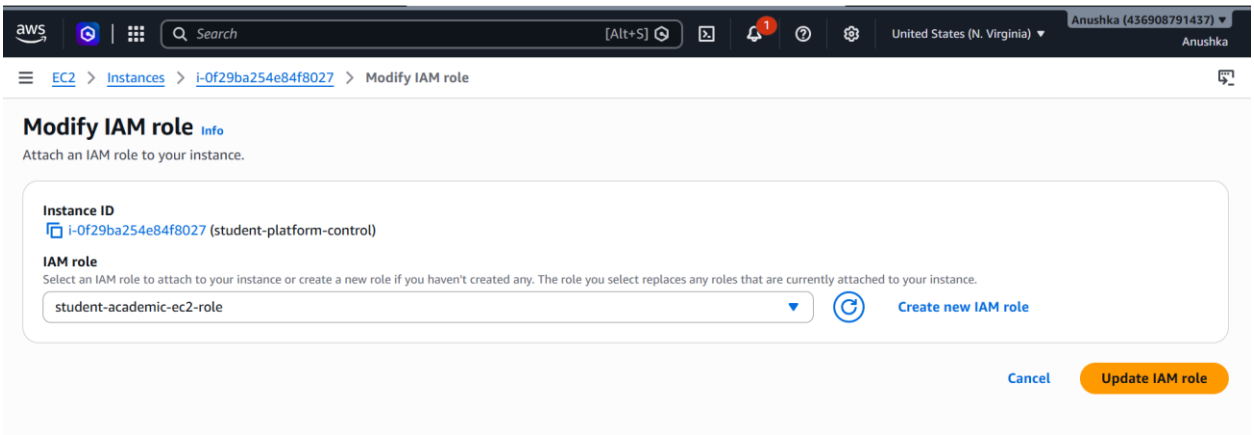
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,. @-/\[{}]!#$%^*():;"'`

Step 2: Add permissions

Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonS3FullAccess	AWS managed	Permissions policy
AmazonVPCFullAccess	AWS managed	Permissions policy
IAMFullAccess	AWS managed	Permissions policy



Check AWS authentication

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ aws sts get-caller-identity
{
    "UserId": "AROAWLOOLC2GVJZGAWBXG:i-0f29ba254e84f8027",
    "Account": "436908791437",
    "Arn": "arn:aws:sts::436908791437:assumed-role/student-academic-ec2-role/i-0f29ba254e84f8027"
}
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Inspect your project

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform
git status
git remote -v
ls -la
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
origin  https://github.com/anushkarajdev2410-netizen/student-academic-platform.git (fetch)
origin  https://github.com/anushkarajdev2410-netizen/student-academic-platform.git (push)
total 60
drwxrwxr-x 8 ubuntu ubuntu 4096 Oct 14 10:33 .
drwxr-x--- 9 ubuntu ubuntu 4096 Oct 14 11:04 ..
-rw-rw-r-- 1 ubuntu ubuntu   75 Oct 14 10:33 .dockerignore
drwxrwxr-x 7 ubuntu ubuntu 4096 Oct 14 11:05 .git
-rw-rw-r-- 1 ubuntu ubuntu  275 Oct 14 10:33 .gitignore
-rw-rw-r-- 1 ubuntu ubuntu  303 Oct 14 10:33 Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 1543 Oct 14 10:33 Jenkinsfile
-rw-rw-r-- 1 ubuntu ubuntu 1164 Oct 14 10:33 README.md
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 14 10:33 app
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 database
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 k8s
-rw-rw-r-- 1 ubuntu ubuntu   58 Oct 14 10:33 requirements.txt
-rw-rw-r-- 1 ubuntu ubuntu  114 Oct 14 10:33 run.py
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 templates
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 10:33 tests
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Check the AWS region

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ aws configure get region
ubuntu@ip-172-31-17-50:~/student-academic-platform$ export AWS_DEFAULT_REGION=us-east-1
```

Create a Terraform directory

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ ls -la
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 14 11:11 .
drwxrwxr-x 9 ubuntu ubuntu 4096 Oct 14 11:11 ..
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$
```

Create the Terraform configuration

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ nano main.tf
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ cat main.tf
terraform {
  required_version = ">= 1.5.0"

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
}

variable "aws_region" {
  description = "AWS deployment region"
  type        = string
  default     = "us-east-1"
}

variable "ssh_allowed_cidr" {
  description = "Your public IPv4 address in CIDR notation, e.g. 203.0.113.10/32"
  type        = string
}

data "aws_ami" "ubuntu" {
  most_recent = true
```

Create the variables file

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ nano terraform.tfvars
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ cat terraform.tfvars
ssh_allowed_cidr = "52.90.161.5/32"
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$
```

Initialize and validate Terraform

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ terraform init
terraform fmt
terraform validate
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.

ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ terraform plan
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 0s [id=ami-0fa5967347d08d2df]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.app will be created
  + resource "aws_instance" "app" {
      + ami                                  = "ami-0fa5967347d08d2df"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
```

Create the infrastructure

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ terraform apply
data.aws_ami.ubuntu: Reading...
data.aws_ami.ubuntu: Read complete after 0s [id=ami-0fa5967347d08d2df]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.app will be created
  + resource "aws_instance" "app" {
      + ami                                  = "ami-0fa5967347d08d2df"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_route_table_association.public: Destroying... [id=rtbassoc-0aa1cf14a78797b6b]
aws_route_table_association.public: Destruction complete after 0s
aws_subnet.public: Destroying... [id=subnet-0ff47c97e1af31b1e]
aws_subnet.public: Destruction complete after 0s
aws_subnet.public: Creating...
aws_subnet.public: Still creating... [00m10s elapsed]
aws_subnet.public: Creation complete after 11s [id=subnet-0b2c1fa3cf1dce147]
aws_route_table_association.public: Creating...
aws_instance.app: Creating...
aws_route_table_association.public: Creation complete after 1s [id=rtbassoc-0cbbe3583d42e3a6b]
aws_instance.app: Still creating... [00m10s elapsed]
aws_instance.app: Creation complete after 13s [id=i-0021f7e46ceb916a5]

Apply complete! Resources: 3 added, 0 changed, 2 destroyed.

Outputs:

instance_id = "i-0021f7e46ceb916a5"
instance_public_ip = "35.153.213.29"
```

Verify the instance

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ terraform output
instance_id = "i-0021f7e46ceb916a5"
instance_public_ip = "35.153.213.29"
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ grep -n -E 'resource "aws_instance"|instance_type|key_name|availability_zone' main.tf
55:  availability_zone      = "us-east-1a"
130:resource "aws_instance" "app" {
132:  instance_type            = "t3.micro"
134:  key_name                 = "student-academic-key"
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ aws ec2 describe-instances \
  --filters "Name=tag:Name,Values=student-academic-app" \
            "Name=instance-state-name,Values=pending,running" \
  --query 'Reservations[].Instances[].[InstanceId,Placement.AvailabilityZone,State.Name,PublicIpAddress]' \
  --output table
-------------------------------------------------------------------
|                       DescribeInstances                         |
+----------------------+-------------+----------+-----------------+
|  i-0021f7e46ceb916a5 |  us-east-1a |  running |  35.153.213.29  |
```

Prepare an SSH key

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ mkdir -p ~/.ssh
chmod 700 ~/.ssh

ssh-keygen -t ed25519 \
  -f ~/.ssh/student-academic-key \
  -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/ubuntu/.ssh/student-academic-key
Your public key has been saved in /home/ubuntu/.ssh/student-academic-key.pub
The key fingerprint is:
SHA256:5/dR6d7Z4KAAw+RPo2imjDF/CQ0DcSl+7xFNu4ESJqg ubuntu@ip-172-31-17-50
The key's randomart image is:
+--[ED25519 256]--+
|o...             |
|+o.o   .         |
|ooo . = .        |
|E.oo * +        .|
|  .+o * S .    ..|
|  . .+ B +    .. |
|o  .=.o o . o o. |
| * +o.   . o +.o+|
|. +.      .   ooo|
+----[SHA256]-----+
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$
```

Check Ansible and project files

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible --version
ansible [core 2.21.5]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.14.4 (main, Aug 20 2026, 10:41:58) [GCC 15.2.0] (/usr/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform/terraform$ cd ~/student-academic-platform

ls -la

find . -maxdepth 3 -type f \
  -not -path './.git/*' \
  -not -path './.venv/*' | sort | head -150
total 64
drwxrwxr-x  9 ubuntu ubuntu 4096 Oct 14 11:11 .
drwxr-x--- 10 ubuntu ubuntu 4096 Oct 14 14:15 ..
-rw-rw-r--  1 ubuntu ubuntu   75 Oct 14 10:33 .dockerignore
drwxrwxr-x  7 ubuntu ubuntu 4096 Oct 14 11:10 .git
-rw-rw-r--  1 ubuntu ubuntu  275 Oct 14 10:33 .gitignore
-rw-rw-r--  1 ubuntu ubuntu  303 Oct 14 10:33 Dockerfile
-rw-rw-r--  1 ubuntu ubuntu 1543 Oct 14 10:33 Jenkinsfile
-rw-rw-r--  1 ubuntu ubuntu 1164 Oct 14 10:33 README.md
drwxrwxr-x  3 ubuntu ubuntu 4096 Oct 14 10:33 app
drwxrwxr-x  2 ubuntu ubuntu 4096 Oct 14 10:33 database
drwxrwxr-x  2 ubuntu ubuntu 4096 Oct 14 10:33 k8s
-rw-rw-r--  1 ubuntu ubuntu   58 Oct 14 10:33 requirements.txt
-rw-rw-r--  1 ubuntu ubuntu  114 Oct 14 10:33 run.py
drwxrwxr-x  2 ubuntu ubuntu 4096 Oct 14 10:33 templates
drwxrwxr-x  3 ubuntu ubuntu 4096 Oct 14 14:10 terraform
drwxrwxr-x  2 ubuntu ubuntu 4096 Oct 14 10:33 tests
./.dockerignore
./.gitignore
./Dockerfile
```

Prepare an Ansible inventory

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ mkdir -p ansible
ubuntu@ip-172-31-17-50:~/student-academic-platform$ nano ansible/inventory.ini
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cat ansible/inventory.ini
[app_servers]
student-app ansible_host=35.153.213.29 ansible_user=ubuntu ansible_ssh_private_key_file=~/.ssh/student-academic-key

[app_servers:vars]
ansible_python_interpreter=/usr/bin/python3
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Test SSH connectivity through Ansible

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ unset SSH_ASKPASS

ansible -i ansible/inventory.ini student_servers -m ping
student-server | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Check the target EC2 server

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible -i ansible/inventory.ini student_servers -m shell -a \
"hostname && lsb_release -a 2>/dev/null | head -n 3 && docker --version 2>/dev/null || true"
student-server | CHANGED | rc=0 >>
ip-172-31-17-50
Distributor ID: Ubuntu
Description:    Ubuntu 26.04 LTS
Release:        26.04
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
```

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ls -la ansible/
total 12
drwxrwxr-x  2 ubuntu ubuntu 4096 Oct 14 16:45 .
drwxrwxr-x 10 ubuntu ubuntu 4096 Oct 14 14:17 ..
-rw-rw-r--  1 ubuntu ubuntu  273 Oct 14 16:45 inventory.ini
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Check your Ansible inventory and application routes

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ^C
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform

cat ansible/inventory.ini

grep -R -nE "@app.route|@.*route|def health|def students|def marks" app tests
[student_servers]
student-server ansible_host=52.90.161.5 ansible_user=ubuntu ansible_ssh_private_key_file=/home/ubuntu/student_platform_key.pem ansible_ssh_common_arg
s='-o IdentitiesOnly=yes -o StrictHostKeyChecking=accept-new' ansible_python_interpreter=/usr/bin/python3
app/routes.py:10:def health():
```

Create the Ansible deployment playbook

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform
nano ansible/site.yml
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cat ansible/site.yml

---
- name: Deploy Student Academic Platform
  hosts: student_servers
  become: true

  vars:
    app_dir: /opt/student-academic-platform
    image_name: student-academic-platform
    container_name: student-academic-platform
    host_port: 8080
    container_port: 5000

  tasks:
    - name: Update apt package cache
      ansible.builtin.apt:
        update_cache: true
        cache_valid_time: 3600

    - name: Install Docker and curl
```

Validate the playbook before deployment

```
        timeout: 60
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform

ansible-playbook -i ansible/inventory.ini ansible/site.yml --syntax-check

playbook: ansible/site.yml
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -m ping
student-server | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Verify the deployment

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible-playbook -i ansible/inventory.ini ansible/site.yml

PLAY [Deploy Student Academic Platform] ****************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [student-server]

TASK [Update apt package cache] ************************************************************************************************************************
ok: [student-server]

TASK [Install Docker and curl] *************************************************************************************************************************
ok: [student-server]

TASK [Start and enable Docker] *************************************************************************************************************************
ok: [student-server]

TASK [Create application directory] ********************************************************************************************************************
changed: [student-server]

TASK [Copy application source] *************************************************************************************************************************
changed: [student-server]

TASK [Copy Dockerfile] *********************************************************************************************************************************
changed: [student-server]

TASK [Copy requirements] *******************************************************************************************************************************
changed: [student-server]

TASK [Copy application entry point] ********************************************************************************************************************
```

```
TASK [Copy application source] ************************************************************************************************************************
changed: [student-server]

TASK [Copy Dockerfile] ********************************************************************************************************************************
changed: [student-server]

TASK [Copy requirements] ******************************************************************************************************************************
changed: [student-server]

TASK [Copy application entry point] *******************************************************************************************************************
changed: [student-server]

TASK [Create persistent database directory] ***********************************************************************************************************
changed: [student-server]

TASK [Build application Docker image] *****************************************************************************************************************
changed: [student-server]

TASK [Remove previous application container if present] ************************************************************************************************
ok: [student-server]

TASK [Run application container] **********************************************************************************************************************
changed: [student-server]

TASK [Wait for Flask application to start] ************************************************************************************************************
ok: [student-server]

PLAY RECAP ********************************************************************************************************************************************
student-server             : ok=14   changed=8    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Check the application response on the target server:

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -b -m shell -a "curl -sS -i http://127.0.0.1:8080/"
student-server | CHANGED | rc=0 >>
HTTP/1.1 200 OK
Server: Werkzeug/3.1.9 Python/3.12.15
Date: Wed, 14 Oct 2026 17:04:08 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 25295
Connection: close


<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Student Academic Management Platform for managing students, faculty, courses and academic performance.">
  <title>CampusFlow | Academic Management</title>
  <style>
    :root {
      --navy: #172554;
      --blue: #4f46e5;
      --blue-light: #eef2ff;
      --bg: #f5f7fb;
      --text: #202b43;
      --muted: #7b879c;
```

Verify the Docker container

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -m ping
student-server | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -b -m command -a "docker ps"
student-server | CHANGED | rc=0 >>
CONTAINER ID   IMAGE                      COMMAND           CREATED         STATUS         PORTS                                         NAMES
084fa0f9e8dc   student-academic-platform   "python run.py"   6 minutes ago   Up 6 minutes   0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp   student-academic-platform
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -b -m command -a "docker logs --tail 50 student-academic-platform"
student-server | CHANGED | rc=0 >>
 * Serving Flask app 'app'
 * Debug mode: offWARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [14/Oct/2026 17:04:08] "GET / HTTP/1.1" 200 -
ubuntu@ip-172-31-17-50:~/student-academic-platform$ ansible student_servers -i ansible/inventory.ini -b -m shell -a "curl -i --max-time 10 http://127.0.0.1:8080/"
student-server | CHANGED | rc=0 >>
HTTP/1.1 200 OK
Server: Werkzeug/3.1.9 Python/3.12.15
Date: Wed, 14 Oct 2026 17:08:43 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 25295
```

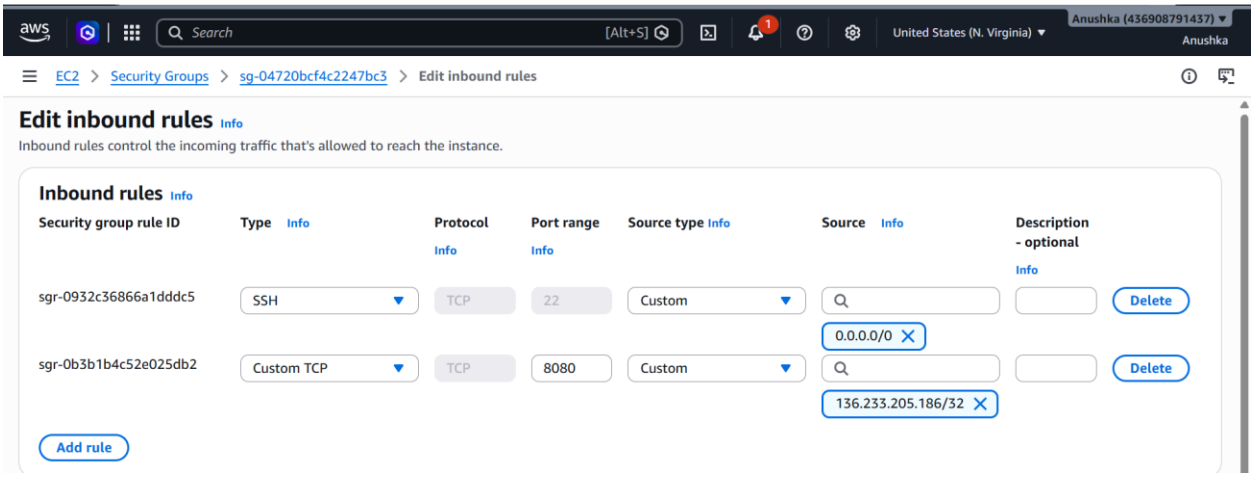
Find the actual API endpoints and inspect tests

```
ubuntu@ip-172-31-17-50:~/student-academic-platform$ cd ~/student-academic-platform

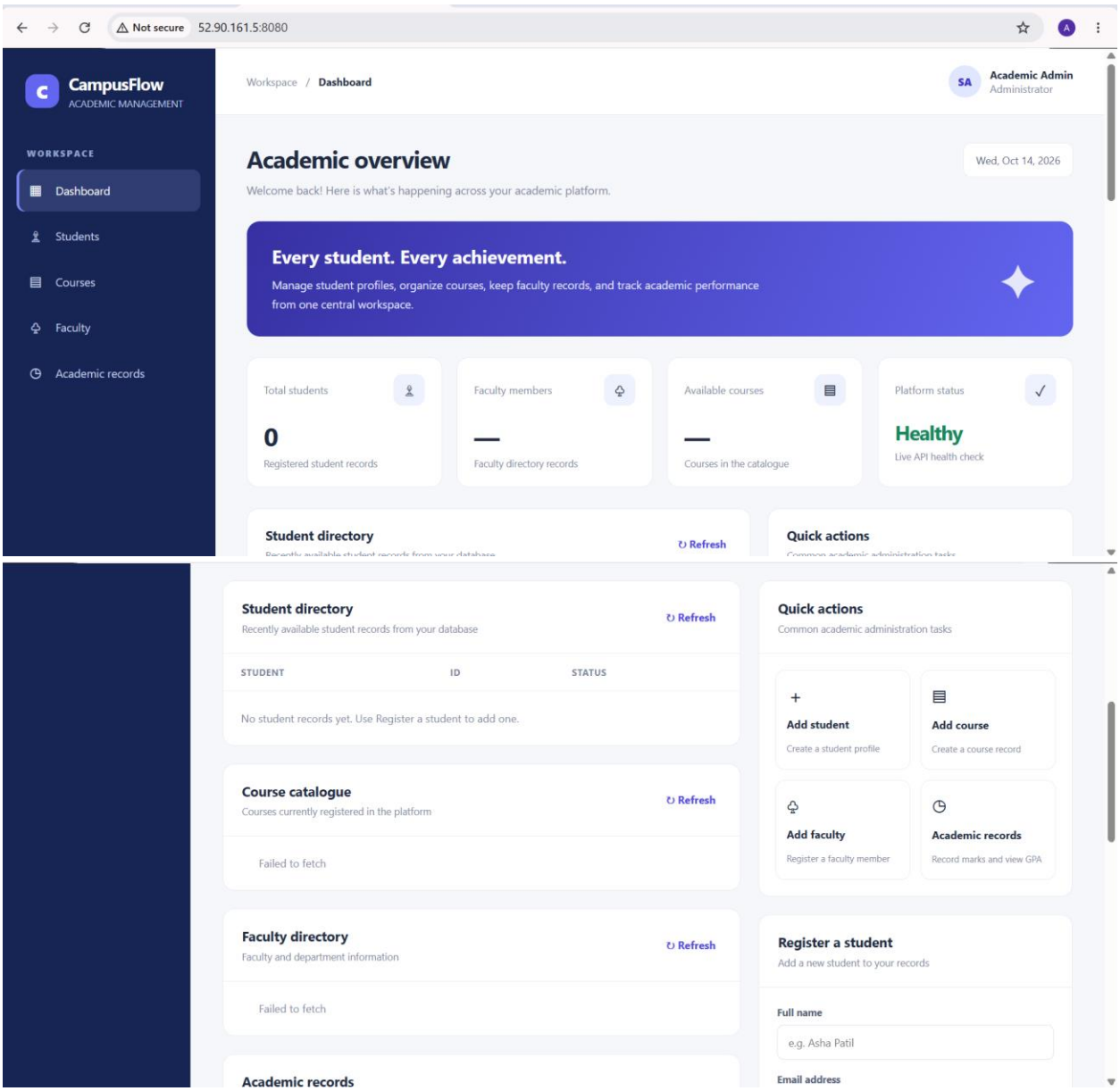
grep -R -nE "@.*route|def health|def students|def marks" app
app/routes.py:10:def health():
ubuntu@ip-172-31-17-50:~/student-academic-platform$ find tests -maxdepth 2 -type f
tests/conftest.py
tests/test_api.py
tests/test_marks_mgmt.py
tests/test_student_service.py
ubuntu@ip-172-31-17-50:~/student-academic-platform$ grep -R -nE "health|students|marks|/api/" tests app
tests/test_api.py:1:def test_health_endpoint(client):
tests/test_api.py:2:    response = client.get("/api/health")
tests/test_api.py:4:    assert response.get_json()["status"] == "healthy"
tests/test_api.py:7:    response = client.post("/api/students", json={
tests/test_api.py:14:    response = client.post("/api/students", json={"name": "", "email": "bad"})
tests/test_marks_mgmt.py:1:from app.services import add_student, add_course, enroll_student, record_marks, calculate_gpa, grade_point
tests/test_marks_mgmt.py:9:def test_marks_and_weighted_gpa(app):
tests/test_marks_mgmt.py:16:        record_marks(sid, c1, "2026-Fall", 95)
tests/test_marks_mgmt.py:17:        record_marks(sid, c2, "2026-Fall", 85)
tests/test_marks_mgmt.py:20:def test_marks_require_enrollment(app):
tests/test_marks_mgmt.py:25:            record_marks(sid, cid, "2026-Fall", 85)
tests/test_student_service.py:7:        row = get_db().execute("SELECT * FROM students WHERE id=?", (student_id,)).fetchone()
app/routes.py:9:@api.get("/health")
app/routes.py:10:def health():
app/routes.py:11:    return jsonify({"status": "healthy", "service": "student-academic-platform"}), 200
app/routes.py:18:            "GET /api/health", "GET/POST /api/students", "GET/POST /api/faculty",
```

Check public access and the AWS security group

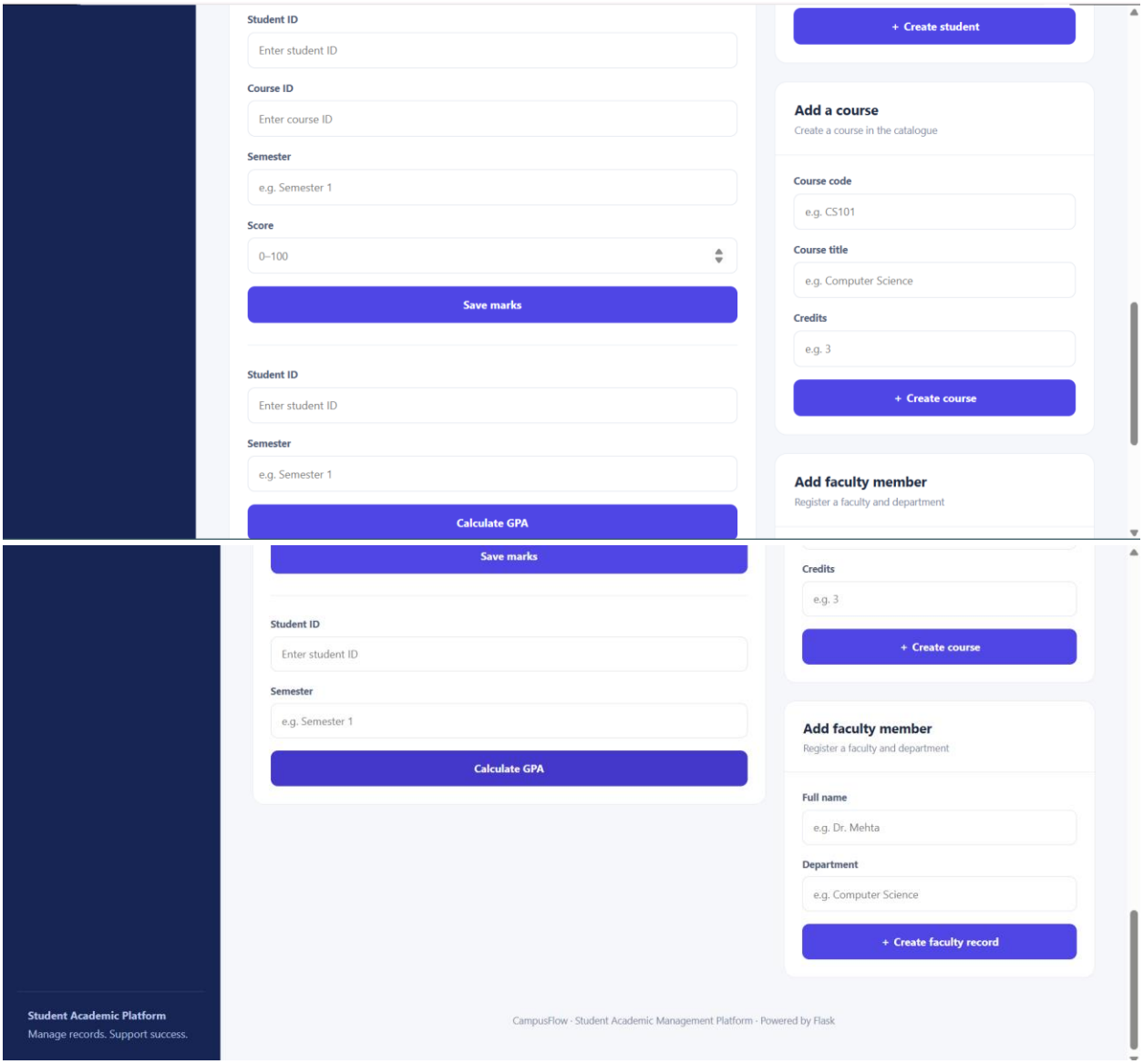
Set Inbound Rules



Check website


Not secure
52.90.161.5:8080
CampusFlow
ACADEMIC MANAGEMENT
WORKSPACE
Dashboard
Students
Courses
Faculty
Academic records
Workspace / Dashboard
SA
Academic Admin
Administrator
Academic overview
Wed, Oct 14, 2026
Welcome back! Here is what's happening across your academic platform.
Every student. Every achievement.
Manage student profiles, organize courses, keep faculty records, and track academic performance from one central workspace.
Total students
0
Registered student records
Faculty members
—
Faculty directory records
Available courses
—
Courses in the catalogue
Platform status
Healthy
Live API health check
Student directory
Refresh
Quick actions

Student directory
Recently available student records from your database
Refresh
STUDENT
ID
STATUS
No student records yet. Use Register a student to add one.
Course catalogue
Courses currently registered in the platform
Refresh
Failed to fetch
Faculty directory
Faculty and department information
Refresh
Failed to fetch
Academic records
Quick actions
Common academic administration tasks
Add student
Create a student profile
Add course
Create a course record
Add faculty
Register a faculty member
Academic records
Record marks and view GPA
Register a student
Add a new student to your records
Full name
e.g. Asha Patil
Email address



Run the project’s test

```
(.venv) ubuntu@ip-172-31-17-50:~/student-academic-platform$ python -m pytest -v
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-8.4.1, pluggy-1.6.0 -- /home/ubuntu/student-academic-platform/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/ubuntu/student-academic-platform
plugins: cov-6.2.1
collected 10 items

tests/test_api.py::test_health_endpoint PASSED                           [ 10%]
tests/test_api.py::test_create_student_endpoint PASSED                   [ 20%]
tests/test_api.py::test_invalid_student_is_rejected PASSED               [ 30%]
tests/test_marks_mgmt.py::test_grade_point_bands PASSED                  [ 40%]
tests/test_marks_mgmt.py::test_marks_and_weighted_gpa PASSED             [ 50%]
tests/test_marks_mgmt.py::test_marks_require_enrollment PASSED           [ 60%]
tests/test_student_service.py::test_student_can_be_created PASSED        [ 70%]
tests/test_student_service.py::test_duplicate_student_email_is_rejected PASSED [ 80%]
tests/test_student_service.py::test_enrollment_cannot_exceed_18_credits PASSED [ 90%]
tests/test_student_service.py::test_same_course_cannot_be_enrolled_twice PASSED [100%]

============================== 10 passed in 0.12s ==============================
```

```
(.venv) ubuntu@ip-172-31-17-50:~/student-academic-platform$ python -m pytest --cov=app --cov-report=term-missing
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-8.4.1, pluggy-1.6.0
rootdir: /home/ubuntu/student-academic-platform
plugins: cov-6.2.1
collected 10 items

tests/test_api.py ...                                                    [ 30%]
tests/test_marks_mgmt.py ...                                             [ 60%]
tests/test_student_service.py ....                                       [100%]

================================ tests coverage ================================
_____________________ coverage: platform linux, python 3.14.4-final-0 _____________________

Name             Stmts   Miss  Cover   Missing
----------------------------------------------
app/__init__.py      15      1    93%   23
app/db.py            26      0   100%
app/routes.py        67     33    51%   15, 26, 39, 43-48, 52, 56-61, 65-72, 76-84, 88-93
app/services.py     110     34    69%   18, 21-27, 31, 34-35, 37, 44-48, 52, 57, 59, 84-85, 87, 102-106, 117, 119, 121, 133, 139, 142, 145
----------------------------------------------
TOTAL               218     68    69%
============================== 10 passed in 0.13s ==============================
(.venv) ubuntu@ip-172-31-17-50:~/student-academic-platform$
```

Stop instance

Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID	Stop protection	Result
i-0f29ba254e84f8027 (student-platform-control)	Disabled	Can stop

Associated resources

You will continue to incur charges for these resources while the instance is stopped

You will be billed for associated resources

After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Skip OS shutdown

This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

Cancel Stop